

Zijian MENG

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Education

- Starting 2026 **Ph.D., Mechanical Engineering and Scientific Computing**
University of Michigan, Ann Arbor, USA
- Sep 2023 – Dec 2025 **Master of Applied Science, Mechanical and Materials Engineering (Co-supervised with Electrical and Computer Engineering)**
Supervised by Laurent Karim Béland and Ryan Grant
Queen's University, Kingston, Canada
4.3/4.3 GPA
- Sep 2018 – May 2023 **Bachelor of Applied Science, Mechanical Engineering**
With Professional Internship
Queen's University, Kingston, Canada
4.18/4.3 GPA

Employment

- Sep 2025 – Dec 2025 **Computational Modelling Intern, Canadian Nuclear Laboratories**, Chalk River, Canada
- Collaborated with experimental scientists to model helium in neutron-irradiated 304 stainless steel from a salvaged component of the National Research Universal reactor.
 - Applied Object Kinetic Monte Carlo simulations for helium and defect distribution and effects.
- Fall 2023 & 2024 **Lead Lab Teaching Assistant, Queen's University**, Kingston, Canada
- Engineering Graphics Course: Led 2 other TAs in lab sessions of more than 80 students.
 - Live demonstrations, management of sessions and students, tutorials, and marking.
- Winter 2023 **Teaching Assistant, Queen's University**, Kingston, Canada
- Automatic Control Course: Communicating with students, marking, and proctoring.
- May 2021 – June 2022 **Product Engineering Intern, Hanon Systems Canada**, Belleville, Canada
- Developed and organized the builds of fluid transport prototypes for automotive thermals.
 - Work included projects for electric vehicles from Ford, GM, and Lucid.
 - Completed 342 prototype build requests under historically high demand.

Research Experience

- Feb 2026 – Present **Research Assistant, Queen's University**, Kingston, Canada
- Nuclear Materials Group, Computational Materials
- Training and running MLIPs for various projects and supporting software advances.
 - Further improved MTP models, yielding up to $8 \times$ faster models with $3 \times$ lower error.

- Sep 2023 – **Master’s Student Researcher**, *Queen’s University*, Kingston, Canada
- Dec 2025 Nuclear Materials Group, Computational Materials
- Development, training, and validation of machine learning interatomic potentials in systems including Na–K–Cl–O, Ni, Si–O, Si–C; involved with Cu–H, Fe–Ni–Cr, Zr–H.
 - Proposed a cost-aware basis set pruning algorithm, optimizing cost-accuracy significantly.
 - Programmed a more optimized CPU and new GPU implementation of the Moment Tensor Potential in LAMMPS.
 - Contributions yielded up to $13 \times$ CPU inference speedups at equal or better accuracy.
- Winter 2023 **Undergraduate Research Project**, *Queen’s University*, Kingston, Canada
- Nuclear Materials Group, Computational Materials
- Development, training, and validation of machine learning interatomic potentials.

Publications

- Accepted 2026 **Meng, Z.**; Zongo, K.; Thoms, M.; Grant, R. E.; Béland, L. K. Accelerating Moment Tensor Potentials through Post-Training Pruning. arXiv preprint arXiv:2510.19737
- Published 2026 **Meng, Z.**; Zongo, K.; Torres, E.; Maxwell, C.; Grant, R.; Béland, L. K. A Kokkos-accelerated moment tensor potential implementation for LAMMPS. *SoftwareX* 2026, 33, 102524.
- Published 2025 **Meng, Z.**; Sun, H.; Torres, E.; Maxwell, C.; Grant, R. E.; Béland, L. K. Small-cell-based fast active learning of machine learning interatomic potentials. *Computational Materials Science* 2025, 256, 113919.

Honors and Awards

- 2026 **Michigan Institute of Computational Discovery and Engineering Fellowship**
- 2024 **UNENE Best Student Presentation**
Best student presentation at the UNENE 2024 conference in the 7-minute category.
- 2024 **NSERC Alexander Bell CGS-M Scholarship**
The Natural Sciences and Engineering Research Council of Canada’s nationwide Master’s graduate scholarship.
- 2023 **R. Samuel McLaughlin Fellowship**
- 2023 **L. M. Arkley Prize**
Awarded for the best fourth-year Mechanical Engineering paper and presentation.
- 2023 **Colin T. Bayne Memorial Award**
Awarded to the graduating Mechanical Engineering student who, in the opinion of the Department, has shown most proficiency in innovative design.
- 2023 **Conn-Gilbert Award**
Awarded to the graduating Mechanical Engineering student with the highest average in core Thermodynamics courses.
- 2023 **3rd Place—Professional Engineers Ontario, Kingston, Engineering Competition**
- Various **Dean’s Scholar**
- 2022 **Lorne C. Elder Scholarship**
- 2021 **Best Project and Best Game Award, CuHacking Hackathon**
- 2018 **Lena MacNeil Scholarship**

Presentations and Posters

- Presentation **Efficiency Advances in Atomistic Modelling of Nuclear Materials.**
2025 UNENE 2025, Technical Presentation Category
- Presentation **Moment Tensor Potential Efficiency Advances**
2025 pyKMC 2025
- Presentation **Simulating Complex Chemistry for Advanced Reactors: Machine Learning Potentials in Action**
2024 UNENE 2024, 7-minute Category
- Poster **Systematic Small-Cell Training Set Selection**
2023 UNENE Student Posters